

Mihir Balsara

617-642-4806 | mihirnbalsara@gmail.com | [linkedin.com/in/mbalsara](https://www.linkedin.com/in/mbalsara) | github.com/GlxryX

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science

- Specialization Threads: Intelligence and Modeling & Simulation

Atlanta, GA

Expected Dec. 2026

EXPERIENCE

Software Engineering Intern

Apple

May 2026 – August 2026

Seattle, WA

- Built a Python/FastAPI service to convert filed Siri bug reports into a runnable test reproducing the failure on real devices.
- Automated device-state setup so each reproduction starts from the exact conditions the bug was reported under.
- Implemented deterministic and LLM-based grading with repeated sampling to separate real failures from flaky results.
- Built a macOS SwiftUI app that replays captured UI recordings on devices to verify fixes without rebuilding the scenario.
- Shipped an on-device session-capture feature, exposing recording over XPC to the desktop tool.

Software Development Intern

PowerPlan

June 10, 2025 – May 2026

Atlanta, GA

- Developed cloud-native enterprise tax/accounting solutions and shared services using .NET Core, C#, and React.
- Implemented microservices architecture with containerization using Docker and orchestration using Kubernetes.
- Collaborated within an Agile Scrum team, utilizing CI/CD and SQL databases to solve high-value business problems.

Cybersecurity Research Intern

Georgia Tech Research Institute

May 2024 – May 2025

Atlanta, GA

- Collaborated with an international team (SSLab, GTRI, KAIST) for the DARPA AIxCC Finals. Qualified for DEFCON 32.
- Developed high-performance testing and data mutation frameworks in Python and C.
- Implemented and managed CI/CD pipelines with Jenkins, automating builds and reducing deployment time by 40%.

Research Assistant

Lincoln Labs

Aug. 2022 – Jul. 2023

Lexington, MA

- Applied data mining and statistical modeling to 500k+ data points, using Python to generate actionable business insights.
- Built automated data visualization pipelines with Matplotlib/Seaborn, reducing report generation time by 60%.
- Authored and published peer-reviewed findings, demonstrating strong analytical and communication skills.

PROJECTS

1Record - 2024 Cornell Health RX Winner | *Python, AI, NLP, Firebase, React*

- Implemented an AI system using Natural Language Processing (LLM) to derive predictive insights from EHR data.
- Developed a full-stack healthcare application using React/TypeScript frontend with a Python and Firebase backend.
- Collaborated in a 48-hour hackathon environment, demonstrating rapid prototyping and problem-solving skills.

Five-Stage Pipelined RISC CPU | *C, Assembly, Digital Logic, Processor Design, Hazard Resolution*

- Implemented a five-stage pipelined RISC CPU in CircuitSim, cutting execution latency through parallelization, a custom forwarding unit, and single-bubble load hazard resolution.
- Achieved a 4.6x latency reduction over non-pipelined baseline.

Spotify Wrapped Mobile App | *Java, Firebase, Machine Learning, REST APIs*

- Integrated web services (Spotify REST API) and machine learning algorithms for personalized music recommendations.
- Developed a cross-platform mobile application with Firebase authentication and real-time data synchronization.
- Implemented a comprehensive unit testing suite with JUnit, achieving 90%+ code coverage for core functionality.

Discord Applications Platform | *JavaScript, Node.js, Distributed Systems, Docker*

- Architected and deployed a scalable distributed system serving 346,075+ users with 99.9% uptime.
- Optimized application performance through profiling and benchmarking, reducing response times by 35%.
- Built a CI/CD pipeline with Jenkins, reducing deployment downtime to under 5 minutes and enabling rapid iterations.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, SQL, C#, JavaScript, TypeScript, Go, Swift, Objective-C

Core Competencies: Distributed Systems, Algorithms, Data Structures, Cloud Computing, Object-Oriented Design, Database Systems, Machine Learning, AI, Big Data, Operating Systems (Linux)

Tools & Platforms: AWS, Docker, Kubernetes, Git, Jenkins, CI/CD, Agile/Scrum, Jest, JUnit

Databases: MongoDB, MySQL, Firebase, Postgres, Oracle DB